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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2025

ME3001-1 – FINITE ELEMENT METHODS
(For ME)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART -A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. The art of subdividing a structure into convenient number of smaller components is known as
 - A) global stiffness matrix
 - B) force vector
 - C) discretization
 - D) convergence
2. The solution obtained by FEM is
 - A) mostly approximate
 - B) always exact
 - C) sometimes exact
 - D) never exact
3. When thin plate is subjected to loading in its own plane only, the condition is called
 - A) Plane Strain
 - B) Plane stress
 - C) Zero strain
 - D) Zero stress
4. For plane strain problems, the following statement is not correct
 - A) normal stress along z direction is zero
 - B) normal strain along z direction is zero
 - C) shear strains along z directions are zero
 - D) both normal and shear strains along z directions are zero
5. The equation for total potential energy of the spring is given by
 - A) Ku
 - B) $Ku+Fu$
 - C) $-Fu + (1/2K\delta^2)$
 - D) $(1/2K\delta^2 + Fu)$
6. Range of poisson's ratio for metals is
 - A) 0.25-1
 - B) 0.45-1
 - C) 0.25-0.50
 - D) 0-0.25
7. The distributed force per unit volume is called
 - A) Axial force
 - B) Body force
 - C) Traction force
 - D) Point load
8. The nature of loading at various locations and other surface conditions are called
 - A) Traction
 - B) Friction
 - C) Surfacing
 - D) Boundary conditions
9. The eight noded quadrilateral element belongs to ----- family of elements
 - A) Lagrange
 - B) Sardipity
 - C) Serendipity
 - D) Interdipity
10. What is the reason for an element in the Serendipity family to have a smaller size of stiffness matrix compare to a similar element in the Lagrange family?
 - A) Absence of interior nodes
 - B) Modified element connectivity
 - C) Lesser interpolation functions
 - D) Irregular arrangements of node
11. In FEA, the solution is said to be converging when
 - A) Element is converging to a point
 - B) Number of elements in increasing
 - C) Residual is trending to zero
 - D) Order of shape function is increasing
12. Shape function is
 - A) Interpolation function
 - B) Iterative function
 - C) Both interpolation and displacement function
 - D) Displacement function



13. The size of the elemental stiffness matrix is equal to the degree of freedom of the
 A) structure B) shape function
 C) beam D) node
14. Example of a 3-D Element is
 A) Bar B) Hexahedron
 C) Triangle D) Rectangle
15. A structure made of several bars, riveted or welded together referred as
 A) bar B) truss
 C) column D) beam
16. Bar and beam elements are considered as a
 A) 2D elements B) 3D elements
 C) Axisymmetric elements D) 1D elements
17. The following shape functions represent degrees of freedom for a beam element at node 1
 A) H_1 and H_2 B) H_2 and H_3
 C) H_1 and H_3 D) H_3 and H_4
18. In cantilever beam the maximum deflection occurs at
 A) Free end B) Through out
 C) Fixed end D) Point of loading
19. For constant strain triangular element, the shape function is
 A) $N_1 + N_2 + N_3 = 2$ B) $N_1 + N_2 + N_3 = 4$
 C) $N_1 + N_2 + N_3 = 1$ D) $N_1 + N_2 + N_3 = 0$
20. Beam element formulation is called
 A) Iso parametric formulation B) Super parametric formulation
 C) Parametric formulation D) Sub parametric formulation

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

- | | Marks | BT* | CO* | PO* |
|--|-------|-----|-----|-----|
| 1. a) Define FEM. With the block diagram explain the general steps involved in FEM. | 08 | L*1 | 01 | 01 |
| b) Derive the equilibrium equations for a 3D elastic body subjected to 3D state of stress. | 08 | L1 | 02 | 02 |
| 2. a) Derive the element stiffness matrix for a bar element by direct method and briefly explain the stages of FEA | 08 | L2 | 01 | 01 |
| b) Explain the principle of minimum potential energy. Using the principle of minimum potential energy, determine the displacements at the nodes for the spring system shown in Fig.Q1 (b). | | | | |

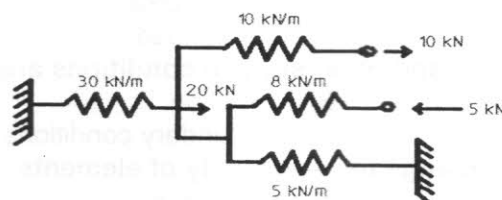


Fig.Q1 (b)

- | | | | | |
|--|----|----|----|----|
| 3. a) What is discretization? Sketch the different types of 1D, 2D, 3D elements used in the finite element analysis. | 08 | L2 | 01 | 01 |
| b) Using Rayleigh-Ritz method, determine the displacement at the midpoint and stresses at the ends of the bar as shown in Fig.Q2 (C) * | | | | |

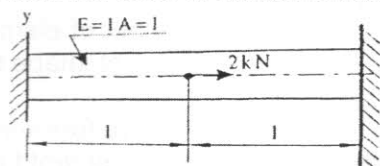


Fig. Q 2 (C)

08	L4	02	02
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Unit – II

4. a) What is convergence in FEM? Explain the p-method and h-method of convergence.
 b) Determine nodal displacements, stress in each material and reaction forces in the bar subjected to concentrated load as shown in Fig. Q 4 (b), using penalty approach of handling boundary condition. Take $E_{Al}=80$ GPa and $E_{St}=210$ GPa.

08 L2 03 01

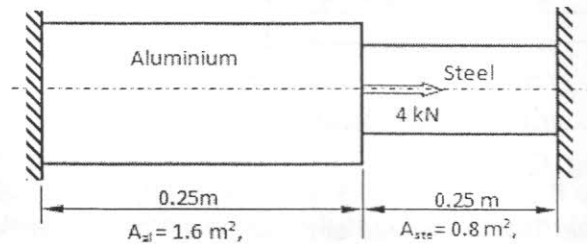


Fig. Q 4(b)

08 L4 04 02

5. a) What are higher order elements? Explain how Pascal's triangle can be used to choose the displacement functions for higher order triangular elements.
 b) A compound bar as shown in Fig. Q5 (b) is subjected to a load of 100kN. Determine nodal displacements, stress in each element and reaction at the support. Take $E_{Cu}=100$ GPa and $E_{St}=200$ GPa.

08 L3 03 01

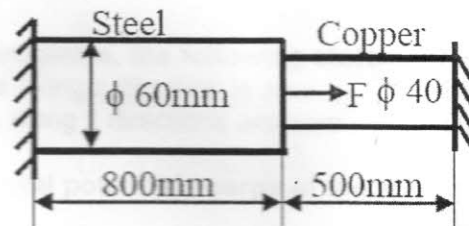


Fig. Q5 (b)

08 L4 04 02

6. a) Derive an expression for shape functions for a vertical 1D quadratic element.
 b) Estimate the nodal displacements and stress in each member for the truss system shown in Fig. Q 6(b). Area of element 1 and 2 is 200 mm^2 , assume $E = 200$ GPa.

08 L3 03 01

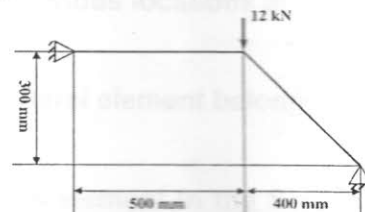


Fig. Q 6(b)

08 L4 04 02

Unit – III

7. Solve for vertical deflection and slopes at points 2 and 3, using beam element for structure shown in Fig.Q7. Also determine the deflection at centre of the portion of beam carrying UDL. Take $E=200 \text{ GPa}$, $I=4 \times 10^6 \text{ mm}^4$

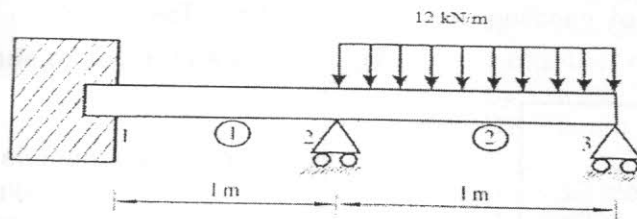


Fig. Q7

8. a) With sketch explain Constant Strain Triangular element.
b) Derive the expression for shape functions for Constant Strain Triangular element.

16 L5 05 02

04 L3 05 01

12 L4 05 02

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
